

Developing Neuro-Symbolic Conversational Systems for Reliable Multi-Perspective Production Process Intelligence

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1 Motivation and goals of the tutorial

Large Language Models (LLMs) have shown impressive capabilities across various process analysis tasks, including process modeling, simulation, and mining [11, 6, 3]. Nonetheless, the extent to which current LLMs can perform genuine logical reasoning remains a critical area of investigation. Although some studies showcase their remarkable performance, the reasoning mechanism in LLMs is fundamentally based on probabilistic pattern recognition rather than formal logic. While they can approximate abstract reasoning patterns, they still fall short of true logical inference [16]. Minor changes in input can significantly affect model outputs, revealing a strong sensitivity to slight variation [13]. Consequently, the likelihood of generating an accurate response quickly decreases with the number of required steps, emphasizing their limited reliability in handling complex reasoning tasks.

This limitation is especially critical in production systems, where operating and interpreting multiple reasoning and process intelligence tools requires highly specialized expertise [4, 10]. For example, revising a production plan to meet delivery deadlines while minimizing disruptions, optimizing resources and inventory, and ensuring system correctness demands complex multi criteria analysis. Today, this work is performed manually by expert users using heterogeneous reasoning and process intelligence tools, making the process time consuming, skill intensive, and prone to suboptimal decisions due to the limited exploration of available data and alternatives [9]. As LLMs are increasingly used to support human operators in these tasks, analytical methods must deliver accurate and reliable diagnostic and prognostic insights within strict time constraints, a guarantee that LLMs alone cannot provide [17].

To address this challenge, in this tutorial *we discuss the required methodological and development steps to create a new breed of trusted neuro-symbolic conversational systems*. These systems integrate LLMs with reasoning engines and process intelligence tools to support accurate, multi perspective analysis of production processes [8]. Rather than adopting an "LLM-as-reasoner" approach, we advocate an "*LLM-as-formalizer*" paradigm [18], in which the LLM

is trained to capture the user’s intent, translate it into machine readable instructions, orchestrate external reasoning and process intelligence tools and generate actionable insights and predictions to analyze and optimize process performance under different scenarios. This hybrid approach ensures that the flexibility of natural language is backed by the sound guarantees of formal solvers, supporting reliability and usability [8]. Through a real world production process use case, we demonstrate how Information Systems researchers and practitioners can design and implement such systems effectively.

2 Outline of the content and learning outcome

This tutorial aims to offer an accessible introduction for students, researchers, and practitioners interested in understanding the logical reasoning mechanisms of Large Language Models (LLMs) and their limitations. Indeed, LLMs still struggle with complex reasoning problems, occasionally deriving unfaithful conclusions that do not follow the reasoning steps prompted by the users [5, 15, 14]. In contrast, reasoning engines that have been developed over several decades by the Artificial Intelligence (AI) community offer mechanisms to guarantee the faithfulness and transparency of their outcomes, a critical aspect for ensuring trusted multi-perspective analysis of production processes.

In this direction, the main learning outcome of this tutorial is to explore, through the development of a conversational system and its implementation on a concrete real-world production process use case, how LLMs can effectively complement reliable state-of-the-art reasoning engines and process intelligence tools to enable accurate and insightful multi-perspective analysis of production processes.

2.1 Background on LLM

First, we provide a concise and self-contained overview of the basic concepts underlying LLMs, covering a spectrum of topics from their differentiation from traditional chatbots and Natural Language Processing (NLP) techniques to their underlying architecture and practical applications in production systems [1].

2.2 Reasoning mechanisms underlying LLMs

Secondly, we present a synthesis of recent advancements in logical reasoning with LLMs, a key area of ongoing AI research. We define the scope of logical reasoning in the context of LLMs, review its theoretical underpinnings, and examine the benchmarks commonly used to assess reasoning capabilities. The tutorial explores current performance across various logical reasoning paradigms, and discusses potential strategies to improve reasoning, including fine-tuning, retrieval-augmented generation, reinforcement learning, and neuro-symbolic methods [12].

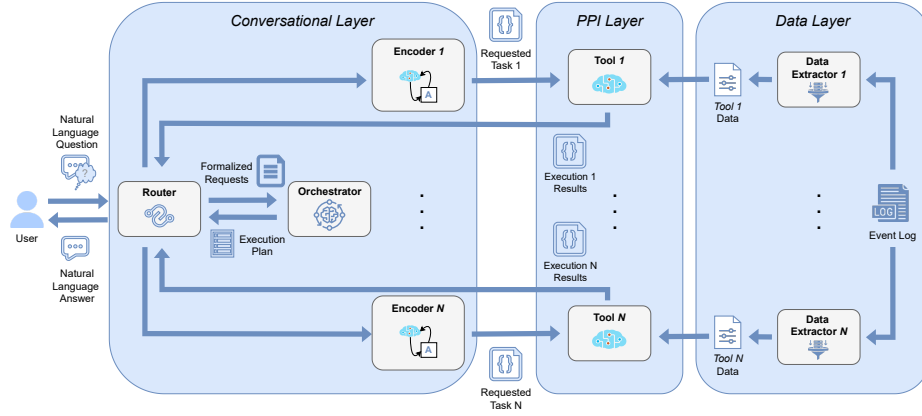


Fig. 1. Domain-agnostic overview of the architecture of a neuro-symbolic conversational system for multi-perspective process intelligence.

2.3 Integrating LLMs with reasoning engines

Specifically, we present a replicable methodology to decompose a reasoning problem into three operational stages: (i) problem formulation, (ii) reasoning, and (iii) results interpretation. During *problem formulation*, the system uses the LLM technology to translate the user’s natural language request into a machine-readable problem compatible with the specific reasoning task to solve (i.e., simulation, formal verification, etc.) and routes it via an orchestrator to the appropriate reasoning engines and process intelligence tools. Subsequently, at the *reasoning* stage, a dedicated reasoning or process intelligence solver processes the machine-readable formulation to obtain an outcome that is translated by an LLM to be human-readable in the *results interpretation* stage.

3 Intended Audience

The tutorial is suitable to researchers, students, practitioners and professionals in Information Systems research. No prerequisite or previous knowledge of LLM is necessary to join the tutorial. During the tutorial, appropriate material will be accessible to the audience, and it will be based on digital handouts, slides, books excerpts, software tools and concrete case studies.

4 Relevance to CAiSE’26 theme and novel aspects

In contrast with recent LLM research in Information Systems, which relies on oversimplified LLM-centric approaches that generate seemingly coherent yet potentially inaccurate and unverifiable explanations for human tasks [2], our vision is more aligned with Kahneman’s dual-process theory of cognition [7]. Indeed,

the novelty lies in combining Conversational AI as System 1 (fast, intuitive, and adaptive), particularly LLMs with their ability to handle natural language ambiguity, with the reasoning capabilities of sound techniques as System 2 (slow, deliberative, and analytical) to ensure transparent and reliable decision-making.

In this regard, the tutorial strongly aligns with this year’s CAiSE theme, “*Rethinking Information System Engineering in a Continuously Changing Information System World*”, by presenting a novel, principled approach for integrating LLMs into information systems that ensures transparent interactions and reliable outcomes.

5 Planned activities to engage the audience

The tutorial is thought to be highly interactive. We will provide to the audience the inputs used to interact with the use case tackled in the range of the tutorial. Moreover, we will make available the last version of the framework implemented for the production process under analysis.

6 Presenters and affiliation

Angelo Casciani is a third-year PhD student in the National PhD in AI at Sapienza University of Rome. He earned his Master’s degree with full marks and honors in Engineering in Computer Science from the same university. His research focuses on the integration neuro-symbolic AI techniques for Business Process Management (BPM), to enable actionable conversations within the vision of AI-augmented BPM Systems. He published 13 papers and serves as reviewer for conferences (CAiSE, ICPM, EDOC, RCIS, and ECAI), journals (ACM Computing Surveys, BISE, DKE, and Information Systems), and is a member of the program committee of ACM SAC (KNLP track).

Livia Lestingi earned a Ph.D. in Information Technology at Politecnico di Milano in 2023, where she is currently a Junior Assistant Professor. Her research interests involve Software Engineering methodologies for the analysis of complex Cyber-Physical Systems, with a focus on automata learning techniques, explainable AI, Federated Learning, and Human-Machine Teaming. She published more than 20 papers and won a Best Presentation Award at the Doctoral Symposium co-located with FM (Formal Methods) in 2021 and a Springer Award for her Ph.D. thesis in 2023. She serves as a reviewer for international journals and is a member of the program committee of international conferences and workshops. She served as the program co-chair of three workshops (FAACS, DataMod, and Ex-ASE) co-located with international conferences in 2024 and 2025.

Andrea Marrella is Associate Professor of Engineering in Computer Science at Sapienza University of Rome. He co-authored over 130 scientific publications in major outlets in the AI and information systems areas, winning a Best Paper Award at CAiSE 2017. Andrea is Associate Editor of the ACM Journal of

Data and Information Quality and ACM Computing Surveys. Among his recent scientific appointments, he was the General Chair of ICPM 2023, and the PC Co-Chair of BPM 2024. In 2023, he co-authored the Manifesto on AI-Augmented BPM, presenting the vision to make processes proactive and explainable through AI. From 2022, he coordinates the working group on AI and BPM of the AIXIA community. In 2024, he organized the Dagstuhl seminar on "*Improving Trust between Humans and Software Robots in Robotic Process Automation*".

Andrea Matta is Professor of Manufacturing at Department of Mechanical Engineering of Politecnico di Milano since 1998. Previously he was Distinguished Professor at Shanghai Jiao Tong University from 2014 to 2016, and visiting scholar at Ecole Centrale Paris, University of California at Berkeley, and Tongji University. He is Editor in Chief of Flexible Services and Manufacturing Journal since 2017. He was awarded with the Shanghai One Thousand Talent and Eastern Scholar.

References

1. Casciani, A., Bernardi, M.L., Cimitile, M., Marrella, A.: Conversational Systems for AI-Augmented Business Process Management. In: Research Challenges in Information Science. Springer (2024)
2. De Luzi, F., Monti, F., Mecella, M.: Engineering Information Systems with Large Language Models. Springer (2025)
3. Estrada-Torres, B., del Río-Ortega, A., Resinas, M.: Mapping the Landscape: Exploring Large Language Model Applications in Business Process Management. In: Enterprise, Business-Process and Inf. Sys. Modeling. Springer (2024)
4. Frigerio, N., Tan, B., Matta, A.: Simultaneous control of multiple machines for energy efficiency: a simulation-based approach. *Int. J. Prod. Res.* **62**(3) (2024)
5. Golovneva, O., Chen, M., Poff, S., Corredor, M., Zettlemoyer, L., Fazel-Zarandi, M., Celikyilmaz, A.: ROSCOE: A suite of metrics for scoring step-by-step reasoning. In: The Eleventh Int. Conf. on Learning Representations, ICLR 2023 (2023)
6. Grohs, M., Abb, L., Elsayed, N., Rehse, J.R.: Large language models can accomplish business process management tasks. In: BPM Workshops (2024)
7. Kahneman, D.: Thinking, fast and slow. Penguin, London (2012)
8. Kambhampati, S., et al.: Position: LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks. In: Forty-first Int. Conf. on ML, ICML 2024 (2024)
9. Keskin, Z., Joosten, D., Klasen, N., Huber, M., Liu, C., Drescher, B., Schmitt, R.H.: LLM-Enhanced Human-Machine Interaction for Adaptive Decision-Making in Dynamic Manufacturing Process Environments. *IEEE Access* **13** (2025)
10. Lestingi, L., et al.: Analyzing the impact of human errors on interactive service robotic scenarios via formal verification. *Softw. Syst. Model.* **23**(2) (2024)
11. Li, J., et al.: Fundamental Capabilities and Applications of Large Language Models: A Survey. *ACM Comput. Surv.* **58**(2) (2026)
12. Liu, H., Fu, Z., Ding, M., Ning, R., Zhang, C., Liu, X., Zhang, Y.: Logical Reasoning in Large Language Models: A Survey (2025), <https://arxiv.org/abs/2502.09100>
13. Lunardi, R., Mea, V.D., Mizzaro, S., Roitero, K.: On Robustness and Reliability of Benchmark-Based Evaluation of LLMs. In: ECAI 2025 (2025)

14. Mirzadeh, I., Alizadeh, K., Shahrokhi, H., Tuzel, O., Bengio, S., Farajtabar, M.: Gsm-symbolic: Understanding the limitations of mathematical reasoning in large language models (2024), <https://arxiv.org/abs/2410.05229>
15. Ribeiro, D.N., Wang, S., Ma, X., Zhu, H., Dong, R., Kong, D., Burger, J., Ramos, A., Huang, Z., Wang, W.Y., Karypis, G., Xiang, B., Roth, D.: STREET: A multi-task structured reasoning and explanation benchmark. In: The Eleventh Int. Conf. on Learning Representations, ICLR 2023 (2023)
16. Song, P., Han, P., Goodman, N.: Large Language Model Reasoning Failures. *Trans. Mach. Learn. Res.* **2026** (2026)
17. Stechly, K., Valmeekam, K., Kambhampati, S.: On the self-verification limitations of large language models on reasoning and planning tasks. In: ICLR 2025 (2025)
18. Tantakoun, M., et al.: LLMs as Planning Formalizers: A Survey for Leveraging Large Language Models to Construct Automated Planning Models. In: ACL (2025)